

Topic VI: Multireference alignment

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1 Background on the DFT

We recall some basic definitions and facts related to the discrete Fourier transform (DFT) that will be essential in what follows. Note that different authors may follow different conventions regarding normalizing constants, but it is easy to convert from one convention to another. For more on the DFT and Fourier analysis in general, see, for example, the reference [5] (among countless others).

We define the Fourier basis vectors of $\mathbb{C}^L$ by

$$\boldsymbol{\varphi}_k[j] = \frac{1}{\sqrt{L}} e^{2\pi i k j / L}, \quad (1)$$

for $k = 0, \dots, L-1$. These basis vectors are orthonormal (exercise). Note too that $\boldsymbol{\varphi}_k = \boldsymbol{\varphi}_{k+L}$.

Then the Fourier transform of a vector $\mathbf{x} \in \mathbb{C}^L$ is defined by

$$(\mathcal{F}\mathbf{x})[k] = \langle \mathbf{x}, \boldsymbol{\varphi}_k \rangle = \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[j] e^{-2\pi i k j / L}. \quad (2)$$

Note that for each j ,

$$\mathbf{x}[j] = \sum_{k=0}^{L-1} \langle \mathbf{x}, \boldsymbol{\varphi}_k \rangle \boldsymbol{\varphi}_k[j] = \frac{1}{\sqrt{L}} \sum_{k=0}^{L-1} (\mathcal{F}\mathbf{x})[k] e^{2\pi i k j / L}. \quad (3)$$

Given $\mathbf{x}$, all coefficients $\mathcal{F}\mathbf{x}[k]$ can be computed at cost $O(L \log(L))$ using the Fast Fourier Transform.

For integer ℓ , define the cyclic translation $\mathbf{x}_\ell[j] = \mathbf{x}[j + \ell]$, where the indices are taken modulo L . Then since the vectors are L -periodic,

$$\begin{aligned} (\mathcal{F}\mathbf{x}_\ell)[k] &= \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x_\ell[j] e^{-2\pi i k j / L} \\ &= \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[j - \ell] e^{-2\pi i k j / L} \\ &= \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[m] e^{-2\pi i k (m - \ell) / L} \\ &= e^{2\pi i k \ell / L} \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[m] e^{-2\pi i k m / L} \\ &= e^{2\pi i k \ell / L} (\mathcal{F}\mathbf{x})[k]. \end{aligned} \quad (4)$$

This is a fundamental property of the DFT; translations get converted into modulations. Another fundamental property

is that if $\mathbf{x}$ is real-valued, then $(\mathcal{F}\mathbf{x})[-k] = (\mathcal{F}\mathbf{x})[k]^*$. To see this, because $\mathbf{x}$ is real, $\mathbf{x}[j] = \mathbf{x}[j]^*$, and so

$$\begin{aligned} (\mathcal{F}\mathbf{x})[-k] &= \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[j] e^{2\pi i k j / L} \\ &= \left(\frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} x[j] e^{-2\pi i k j / L} \right)^* \\ &= (\mathcal{F}\mathbf{x})[k]^*, \end{aligned} \tag{5}$$

as claimed.

2 The multireference alignment problem

Let $\mathbf{x} \in \mathbb{R}^L$ be an unknown vector. We observe n observations $\mathbf{y}_1, \dots, \mathbf{y}_n$, each of the form

$$\mathbf{y}_m[j] = \mathbf{x}_{r_m}[j] + \varepsilon_m, \tag{6}$$

where the ε_m are vectors of noise with $\varepsilon_m[j]$ independent $\mathcal{N}(0, L\sigma^2)$, and the r_m are *unknown* translations; $\mathbf{x}_{r_m}[j] = \mathbf{x}[j+r_m]$. The goal is to estimate $\mathbf{x}$, up to a global translation. This is called the *multireference alignment problem* [3, 4].

If the noise level σ is small, then one can *align* the observations, and then average them to remove the noise, with an error of order $O(\sigma/\sqrt{n})$ (with constants depending on L , which we ignore because we treat L as fixed).

However, if σ is large, alignment provably fails; basically, one ends up just aligning noise [2]. It may not be a priori clear that denoising is possible in this case; however, it turns out that it is still possible.

It is easier to work in the Fourier domain; note that recovering the Fourier transform of $\mathbf{x}$ is equivalent to recovering $\mathbf{x}$ itself.

Taking Fourier transforms,

$$(\mathcal{F}\mathbf{y}_m)[k] = (\mathcal{F}\mathbf{x}_{r_m})[k] + (\mathcal{F}\varepsilon_m)[k] \equiv (\mathcal{F}\mathbf{x}_{r_m})[k] + \boldsymbol{\xi}_m[k] \tag{7}$$

Note that the noise vectors $\boldsymbol{\xi}_m$ have mean 0, and for all k, ℓ ,

$$\begin{aligned} \mathbb{E} [\boldsymbol{\xi}_m[k] \overline{\boldsymbol{\xi}_m[\ell]}] &= \frac{1}{L^2} \sum_{j=0}^{L-1} \sum_{j'=0}^{L-1} e^{-2\pi i j k / L} e^{2\pi i j' \ell / L} \mathbb{E} [\varepsilon_m[j] \varepsilon_m[j']] \\ &= \frac{1}{L^2} \sum_{j=0}^{L-1} \sum_{j'=0}^{L-1} e^{-2\pi i j k / L} e^{2\pi i j' \ell / L} \sigma^2 L \delta_{j=j'} \\ &= \frac{\sigma^2}{L} \sum_{j=0}^{L-1} e^{2\pi i j (\ell - k) / L} \\ &= \sigma^2 \delta_{\ell=k}. \end{aligned} \tag{8}$$

So the problem is equivalent to observing the n vectors $\mathcal{F}\mathbf{y}_1, \dots, \mathcal{F}\mathbf{y}_n$ of the form $(\mathcal{F}\mathbf{y}_m)[k] = (\mathcal{F}\mathbf{x})[k] e^{2\pi i r_m / L} + \boldsymbol{\xi}_m[k]$, where the r_m are unknown translations and the $\boldsymbol{\xi}_m$ are independent complex Gaussian vectors with entry-wise variances σ^2 . We will assume for simplicity that $(\mathcal{F}\mathbf{x})[k] \neq 0$ for all k .

To recover $\mathcal{F}\mathbf{x}$ (up to a global phase shift) from these measurements, we will define several *invariants* of the signal (that is, quantities invariant to translations).

First is the mean, i.e. $(\mathcal{F}\mathbf{x})[0]$.

Second is the power spectrum:

$$\mathcal{P}_{\mathbf{x}}[k] = |\mathcal{F}\mathbf{x}[k]|^2. \tag{9}$$

Third is the bispectrum:

$$\mathcal{B}_{\mathbf{x}}[k, \ell] = (\mathcal{F}\mathbf{x})[k] (\mathcal{F}\mathbf{x})[\ell]^* (\mathcal{F}\mathbf{x})[\ell - k]. \tag{10}$$

All these quantities are identical for $\mathbf{x}$ and $\mathbf{x}_r$, for any translate r ; this can be easily checked using the properties we have derived of the DFT. Furthermore, it not hard to see that all three quantities can be consistently estimates from the n observations $\mathbf{y}_1, \dots, \mathbf{y}_n$ in the limit $n \rightarrow \infty$. Finally, it can be shown that $\mathbf{x}$ can be recovered up to a global translation from these invariants.

Estimating the mean $(\mathcal{F}\mathbf{x})[0]$ is easy, just by averaging $(\mathcal{F}\mathbf{y}_m)[0]$ for $m = 1, \dots, n$:

$$\frac{1}{n} \sum_{m=1}^n (\mathcal{F}\mathbf{y}_m)[0] = (\mathcal{F}\mathbf{x})[0] + \frac{1}{n} \sum_{m=1}^n (\mathcal{F}\boldsymbol{\xi}_m)[0] \approx (\mathcal{F}\mathbf{x})[0], \quad (11)$$

the error decays like $O(\sigma/\sqrt{n})$ (with a constant depending on L , that we will ignore since we treat L as fixed).

Similarly, the power spectrum may be estimated by first averaging the power spectra of the observations, and the debiasing to remove the noise variance:

$$\begin{aligned} \frac{1}{n} \sum_{m=1}^n \mathcal{P}_{\mathbf{y}_m}[k] &= \frac{1}{n} \sum_{m=1}^n |(\mathcal{F}\mathbf{y}_m)[k]|^2 \\ &= |(\mathcal{F}\mathbf{x})[k]|^2 + \frac{1}{n} \sum_{m=1}^n |(\mathcal{F}\boldsymbol{\xi}_m)[k]|^2 + \frac{2}{n} \sum_{m=1}^n \langle (\mathcal{F}\mathbf{x})[k], (\mathcal{F}\boldsymbol{\xi}_m)[k] \rangle \\ &\approx |(\mathcal{F}\mathbf{x})[k]|^2 + \sigma^2, \end{aligned} \quad (12)$$

with an error that decays like $O(\sigma^2/\sqrt{n})$.

The bispectrum of $\mathbf{x}$ may be estimated in a similar way, by averaging the bispectra of the observed vectors and debiasing the appropriate terms. Details are left as an exercise. Because each term of the bispectrum is a product of three Fourier coefficients, the estimation error decays like $O(\sigma^3/\sqrt{n})$.

2.1 Phase retrieval

Because we can estimate the power spectrum of $\mathbf{x}$, which we assume is non-vanishing, we can normalize each $(\mathcal{F}\mathbf{x})[k]$ by $\sqrt{\mathcal{P}_{\mathbf{x}}[k]}$, and assume from the outset that $|(\mathcal{F}\mathbf{x})[k]| = 1$ for all k . The problem then reduces to finding the phases of each Fourier coefficient, a well-known problem called *phase retrieval*.

2.2 Frequency marching

Assume we know $\mathcal{B}_{\mathbf{x}}$ exactly. The first step is to find the L possible values of $(\mathcal{F}\mathbf{x})[1]$, which we do following the approach of [3]:

$$\begin{aligned} &\mathcal{B}_{\mathbf{x}}[1, 2]\mathcal{B}_{\mathbf{x}}[2, 3]\mathcal{B}_{\mathbf{x}}[3, 4]\dots\mathcal{B}_{\mathbf{x}}[L-1, L] \\ &= [(\mathcal{F}\mathbf{x})[1](\mathcal{F}\mathbf{x})[2]^*(\mathcal{F}\mathbf{x})[1]] \cdot [(\mathcal{F}\mathbf{x})[2](\mathcal{F}\mathbf{x})[3]^*(\mathcal{F}\mathbf{x})[2]] \cdot [(\mathcal{F}\mathbf{x})[3](\mathcal{F}\mathbf{x})[4]^*(\mathcal{F}\mathbf{x})[3]] \cdots [(\mathcal{F}\mathbf{x})[L-1](\mathcal{F}\mathbf{x})[L]^*(\mathcal{F}\mathbf{x})[L-1]] \\ &= (\mathcal{F}\mathbf{x})[1]^L, \end{aligned} \quad (13)$$

where we use that $(\mathcal{F}\mathbf{x}[k])(\mathcal{F}\mathbf{x}[k])^* = |(\mathcal{F}\mathbf{x}[k])|^2 = 1$, and that $(\mathcal{F}\mathbf{x})[L] = (\mathcal{F}\mathbf{x})[0]$ is known.

We can then pick any of the L roots of $(\mathcal{F}\mathbf{x})[1]^L$ and declare that to be $(\mathcal{F}\mathbf{x})[1]$, since each one corresponds to a different cyclic translation of $\mathbf{x}$, all of which are equally valid.

With our choice of $(\mathcal{F}\mathbf{x})[1]$, we can perform *frequency marching*, using that

$$\mathcal{B}_{\mathbf{x}}[k, k+1] = \mathcal{B}_{\mathbf{x}}[k]\mathcal{B}_{\mathbf{x}}[k+1]^*\mathcal{B}_{\mathbf{x}}[1] \quad (14)$$

allows us to solve for $\mathcal{B}_{\mathbf{x}}[k+1]$ in terms of $\mathcal{B}_{\mathbf{x}}[k]$.

This proves that there is, in principle, enough information to recover $\mathbf{x}$ up to a global translation from the observed data.

3 Aperiodic MRA

Now suppose that the r_m are drawn iid from some unknown probability distribution $\boldsymbol{\rho}$ over $\{0, \dots, L-1\}$. Then it turns out that so long as the distribution $\boldsymbol{\rho}$ has minimal period L , then recovery is possible from the second moment matrix of the

observations, which is estimable with error of order $O(\sigma^2/\sqrt{n})$, not $O(\sigma^3/\sqrt{n})$. This was proven in [1], and we follow the spectral algorithm given there.

First, observe that, for all k , the first moment is

$$\frac{1}{n} \sum_{m=1}^n (\mathcal{F}\mathbf{x}_{r_m})[k] \approx \sum_{m=0}^{L-1} \boldsymbol{\rho}[m](\mathcal{F}\mathbf{x}_m)[k] = \sum_{m=0}^{L-1} \boldsymbol{\rho}[m](\mathcal{F}\mathbf{x})[k] e^{2\pi i m k/L} = (\mathcal{F}\mathbf{x})[k](\mathcal{F}\boldsymbol{\rho})[k]^* \equiv \mathcal{M}_1[k], \quad (15)$$

and so

$$\frac{1}{n} \sum_{m=1}^n (\mathcal{F}\mathbf{y}_m)[k] = \frac{1}{n} \sum_{m=1}^n (\mathcal{F}\mathbf{x}_{r_m})[k] + \frac{1}{n} \sum_{m=1}^n (\mathcal{F}\boldsymbol{\xi}_m)[k] \approx \frac{1}{n} \sum_{m=1}^n (\mathcal{F}\mathbf{x}_{r_m})[k] \quad (16)$$

with error decaying like $O(\sigma/\sqrt{n})$.

As for the second moment, for indices k and ℓ ,

$$\begin{aligned} \frac{1}{n} \sum_{m=1}^n \mathcal{F}\mathbf{x}_{r_m}[k] \mathcal{F}\mathbf{x}_{r_m}[\ell]^* &\approx \sum_{m=0}^{L-1} \boldsymbol{\rho}[m] \mathcal{F}\mathbf{x}_m[k] \mathcal{F}\mathbf{x}_m[\ell]^* \\ &= \sum_{m=0}^{L-1} \boldsymbol{\rho}[m](\mathcal{F}\mathbf{x})[k] e^{2\pi i m k/L} (\mathcal{F}\mathbf{x})[\ell]^* e^{-2\pi i m \ell/L} \\ &= (\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \sum_{m=0}^{L-1} \boldsymbol{\rho}[m] e^{2\pi i m k/L} e^{-2\pi i m \ell/L} \\ &= (\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \sum_{m=0}^{L-1} \boldsymbol{\rho}[m] e^{2\pi i m(k-\ell)/L} \\ &= (\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \boldsymbol{\rho}[\ell - k] \equiv \mathcal{M}_2[k, \ell]. \end{aligned} \quad (17)$$

This can be estimated by using

$$\frac{1}{n} \sum_{m=1}^n \mathcal{F}\mathbf{y}_{r_m}[k] \mathcal{F}\mathbf{y}_{r_m}[\ell]^* \approx (\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \boldsymbol{\rho}[\ell - k] + \frac{1}{n} \sum_{m=1}^n \boldsymbol{\xi}_m[k] \boldsymbol{\xi}_m[\ell]^* \approx (\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \boldsymbol{\rho}[\ell - k] + \sigma^2 \delta_{k=\ell}, \quad (18)$$

and debiasing the diagonal elements by subtracting σ^2 . The error of estimation is of order $O(\sigma^2/\sqrt{n})$.

It is then easy to see that the vectors $\mathcal{F}\mathbf{x}_m$ are eigenvectors of the second moment matrix, with corresponding eigenvalues $\boldsymbol{\rho}[m]$, $m = 0, \dots, L-1$.

Indeed, for all k ,

$$\begin{aligned} \sum_{\ell=0}^{L-1} \mathcal{M}_2[k, \ell](\mathcal{F}\mathbf{x}_m)[\ell] &= \sum_{\ell=0}^{L-1} \mathcal{M}_2[k, \ell](\mathcal{F}\mathbf{x})[\ell] e^{2\pi i m \ell/L} \\ &= \sum_{\ell=0}^{L-1} \left[(\mathcal{F}\mathbf{x})[k](\mathcal{F}\mathbf{x})[\ell]^* \boldsymbol{\rho}[\ell - k] \right] (\mathcal{F}\mathbf{x})[\ell] e^{2\pi i m \ell/L} \\ &= \sum_{\ell=0}^{L-1} (\mathcal{F}\mathbf{x})[k](\mathcal{F}\boldsymbol{\rho})[\ell - k] e^{2\pi i m \ell/L} |(\mathcal{F}\mathbf{x})[\ell]|^2 \\ &= (\mathcal{F}\mathbf{x})[k] \sum_{\ell=0}^{L-1} (\mathcal{F}\boldsymbol{\rho})[\ell - k] e^{2\pi i m \ell/L} \\ &= (\mathcal{F}\mathbf{x})[k] \sum_{\ell=0}^{L-1} (\mathcal{F}\boldsymbol{\rho})[\ell] e^{2\pi i m(\ell+k)/L} \\ &= (\mathcal{F}\mathbf{x})[k] e^{2\pi i m k/L} \sum_{\ell=0}^{L-1} (\mathcal{F}\boldsymbol{\rho})[\ell] e^{2\pi i m \ell/L} \\ &= (\mathcal{F}\mathbf{x})[k] e^{2\pi i m k/L} \boldsymbol{\rho}[m], \end{aligned} \quad (19)$$

as claimed.

Consequently, so long as there is at least one distinct value of $\rho[m]$, different from the others, the corresponding eigenvector is the vector of phases of $\mathcal{F}\mathbf{x}$, which solves the phase retrieval problem.

However, there may be distributions ρ for which there is no entry that is distinct from the others. We can rectify this problem by performing a pre-processing step on the data. This process is randomized, but it works with probability 1 if the distribution ρ has minimal period L , i.e. is aperiodic.

First, draw a distribution θ , from some continuous distribution over the space of probability distributions over $\{0, \dots, L-1\}$.

Then draw n independent samples from θ , $s_1, \dots, s_n$. Cyclically rotate each observation by s_j . Then observe that the distribution of the noise does not change, but for each observation the rotated vector is now $\mathbf{x}_{r_m+s_m}$, $m = 1, \dots, n$.

The distribution of the new rotations $q = r + s$ is

$$\mathbb{P}(q) = \sum_s \rho(q-s)\theta[s] = (\rho * \theta)[q], \quad (20)$$

the convolution of ρ and θ .

We will then show that all entries of the convolution $\rho * \theta$ are distinct, with probability 1; by the union bound, it is enough to show that any two entries i and j are distinct with probability 1.

We have $(\rho * \theta)[i] = \mathbf{e}_i^T \mathbf{C}_\rho \theta$ for all i , so that if $(\rho * \theta)[i] = (\rho * \theta)[j]$, then $(\mathbf{e}_i - \mathbf{e}_j)^T \mathbf{C}_\rho \theta = 0$; but by the randomness of θ , this occurs with probability 0 unless $(\mathbf{e}_i - \mathbf{e}_j)^T \mathbf{C}_\rho \neq 0$, or equivalently, if ρ is equal to a translation of itself by $i - j$, contradicting aperiodicity.

References

- [1] Emmanuel Abbe, Tamir Bendory, William Leeb, João M. Pereira, Nir Sharon, and Amit Singer. Multireference alignment is easier with an aperiodic translation distribution. *IEEE Transactions on Information Theory*, 65(6):3565–3584, 2018.
- [2] Cecilia Aguerrebere, Mauricio Delbracio, Alberto Bartesaghi, and Guillermo Sapiro. Fundamental limits in multi-image alignment. *IEEE Transactions on Signal Processing*, 64(21):5707–5722, 2016.
- [3] Tamir Bendory, Nicolas Boumal, Chao Ma, Zhizhen Zhao, and Amit Singer. Bispectrum inversion with application to multireference alignment. *IEEE Transactions on Signal Processing*, 66(4):1037–1050, 2017.
- [4] Amelia Perry, Jonathan Weed, Afonso Bandeira, Philippe Rigollet, and Amit Singer. The sample complexity of multi-reference alignment. *arXiv preprint at arXiv:1707.00943*, 2017.
- [5] Elias M. Stein and Rami Shakarchi. *Fourier Analysis: An Introduction*. Princeton University Press, 2003.